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RESEARCH ARTICLE

Advancement of information extraction use in legal documents

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ABSTRACT

The number of legal documents is already huge and growth is occurring at a swift pace; yet, the need to procure information from these documents for various purposes persists. One of the technologies available for this purpose is information extraction (IE), which has been applied widely in different domains to date. However, few research reviews have specifically discussed the development of legal IE. For this reason, it is considered important to conduct a literature review to identify and analyse the research, topics, datasets and methods used in legal IE research. This study used a standard systematic review method as stipulated by the PRISMA guidelines and collected papers from five main databases published between January 1990 and December 2020. A total of 107 articles were selected for inclusion in this review. Legal IE has been employed for various applications to date, but there is a wide range of different conditions, problems, and solutions reported by various studies in applying legal IE in various countries. This study offers a more concrete understanding, a broader perspective and provides an overview of various open problems in this regard so that it can serve as a guide in the development of future legal IE research.

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KEYWORDS
Information extraction; legal document; legal information extraction; literature review; systematic review

1. Introduction

The number of legal documents is already vast, with growth only continuing from this point onward. On the other hand, there is a need to be able to process information from these documents for various purposes, such as legal case mapping based on the location of the incident and the articles of indictment law, reviewing the history of similarity in legal cases, building legal hierarchies and analysing data on contract documents. One of the concepts that can be applied to the work of extracting information in documents is IE.

IE is part of natural language processing (NLP), a kind of artificial intelligence able to extract important facts from documents about types of events, various entities or relationships between previously defined entities. IE creates semantic content representations

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that have more valuable meaning, which is used to provide structured input from the extraction results that have high pattern complexity (Grishman 1997; Riloff 1983).

Research on the use of IE has been carried out in various domains to date, including that performed to study rules (rule learner) for extracting information from semi-structured web documents by Jiménez and Corchuelo in 2016 (Jiménez and Corchuelo 2016). Meanwhile, research seeking to extract professional data details such as names, e-mails, addresses, contact numbers and specialities from doctors' websites using the DeepDive was developed by Stanford in 2018 (Vyas et al. 2018).

Some research on legal IE, such as that by Cardellino et al. in 2017, focuses on the extraction of information in legal texts by establishing entity identifiers, classifiers and liaisons so that relevant parts of the text can be identified and correlated with structured knowledge representations using LKIF Core Ontology (Cardellino et al. 2017). Other work by Ii, Katz and Detterman in 2018 focused on NLP and the extraction of information for regulatory and legal documents (Bommarito, Katz, and Detterman 2018). Meanwhile, efforts to extract information from statutory documents, which employed a system for extracting information on the provisions of penalties from Indonesian legal documents, used the ripple-down rule learner to find the legal hierarchy in work by Hartadi and Budi in 2017 (Hartadi and Budi 2018). Finally, Solihin and Budi in 2018 researched the extraction of information from general criminal court decision documents using the rule-based method (Solihin and Budi 2019).

From this range of prior work, it can be seen that legal IE research is very diverse in terms of problems, the use of source documents, the methods employed, the entity being searched for and the stages carried out. This variety can be considered as the primary driving factor for why this systematic review, which aimed to construct a concrete picture of views and understandings of various matters in legal IE research and to provide recommendations on open issues that can be used as guidelines in developing future legal IE research, was pursued.

Specifically, the purpose of this research was to discern (1) what is the state of the development of research in legal IE, (2) what topics of research are selected by researchers in legal IE, (3) which type of legal datasets is the most frequently used one for legal IE, (4) what kinds of methods are used for legal IE and (5) what are the open problems and ongoing developments in legal IE research? This review assessed research conducted between January 1990 and December 2020.

In this report, an explanation of the methodology is presented in section 2 and the results are shared in section 3. Section 4 contains a discussion of the findings and open issues, while the conclusions of this study are shown in section 5.

2. Methods

A systematic review (SR) is a process for identifying, evaluating analysing and summarising all research sources in detail and comprehensively based on a set of systematic research questions (Khan et al. 2003; Kitchenham 2004). This study employs a Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA)-compliant standard SR method (Moher et al. 2016). In addition, several styles of presentation of the review results were also motivated by prior systematic review papers by Romi (Wahono 2007) and D. Claro et al. (2019).

The three main stages conducted in this SR were planning, implementation and reporting. The planning stage, which involves the identification of the need for a review, was discussed in the introduction (section 1) of this paper. The implementation or development of a review protocol was designed to guide studies and reduce researcher bias, was discussed in the methods (section 2) of this paper. This development of a review protocol have the following steps: defining the research question, the strategy in the search process, the inclusion and exclusion criteria for the selection process, the quality assessment process and the process for extracting and synthesising data.

2.1. Research questions

The purpose of identifying research is to locate as many significant studies as possible related to the research question using a search strategy so that the search results are not biased. In this study, the search process was carried out several times using various combinations of search terms.

There were five research questions used in this study, namely:

- RQ1. What is the state of the development of research in legal IE?
- RQ2. What topics of research are selected by researchers in legal IE?
- RQ3. Which type of legal datasets is the most frequently used one for legal IE?
- RQ4. What kinds of methods are used for legal IE?
- RQ5. What are the open problems and ongoing developments in legal IE research?

Concerning RQ1, this study tries to consider information and reviews in terms of the number of papers (journal or proceeding conference) published per year, the contributions of individual researchers, the organisations and countries with which the researchers are affiliated and the states in which the legal documents were used as research objects. Eligible articles for inclusion in this study were those published from 1990 to 2020 – specifically, during a period of 30 years from January 1990 to December 2020.

On RQ2, this investigation chose a scope related to the research questions and research topics of the selected studies. For RQ3, this study examined the various types of datasets, languages and quantities used during the research. Meanwhile, RQ4 focused the review in terms of the methods and techniques used in legal IE. Finally, RQ5 considers an open problem for the development of legal IE research in the future. The motivation behinds each research question can be seen in Table 1.

2.2. Search process

The selection of studies involved choosing digital libraries, specifying search strings, obtaining suitable search strings by repeating tests and refining the search strings. The success of this step relies on an early list of studies from the research database that matches the search keyword.

The literature database used in this study was chosen based on the recommendations of Brereton et al. (2007) who identified electronic sources relevant to software

Table 1. Research Questions and Corresponding Goals During the Literature Review.

ID	Research Question	Goal
RQ1	What is the state of the development of research in legal IE?	Recognise the number of papers that discuss legal IE and recognise the most significant journal, conference and active researchers in legal IE
RQ2	What topics of research are selected by researchers in legal IE?	Recognise topics of research and trends in legal IE
RQ3	Which type of legal datasets is the most frequently used one for legal IE?	Recognise datasets frequently used in legal IE
RQ4	What kinds of methods are used for legal IE?	Recognise the most frequently used methods for legal IE and recognise the proposed methods for improving the legal IE process
RQ5	What are the open problems and ongoing developments in legal IE research?	Recognise an open problem for the development of legal IE research in the future

engineering and considered the claims of several databases. The five research databases used in this SR are as follows:

- Association of Computing Machinery (ACM) Digital Library[1]
- Institute of Electrical and Electronics Engineers (IEEE) eXplore[2]
- Scopus[3]
- SpringerLink[4]
- ScienceDirect[5]

Search keywords were created using several development steps, namely:

(1) Recognise search terms, synonyms and alternative spellings of the research question.
(2) Recognise search terms from titles, abstracts and keywords that are relevant to legal IE.
(3) Create a search string and use Boolean operators such as AND and OR and use double quotation marks to ensure the phrases' order is correct.

The main purpose of this SR was to review the application of information extraction in the legal sector. From this central point, we obtained the term 'information extraction' and related legal search strings. The string 'information extraction' was developed by adding 'relation extraction' and 'named entity recognition', which are part of 'information extraction'. Meanwhile, the string 'legal' were added with 'law' and 'court' as a search string expansion. All these strings were then chained with Boolean operators as follows: *('information extraction' OR 'relation extraction' OR "named entity recognition" OR 'event extraction' OR 'reference resolution') AND (law OR legal OR court)*

Each database has different query-writing standards, so it was necessary to create specific queries for each database. The query used for each database can be seen in Table 2.

The search string obtained was then adjusted to the rules for writing the search string from each database. Search results were limited to journal papers and conference proceedings published in English between January 1990 and December 2020.

Table 2. The Query Strings Used in Each Database.

Database	Query
ACM Digital Library	[Full Text: "information extraction"] OR [Full Text: "relation extraction"] OR [Full Text: "named entity recognition"] OR [Full Text: "event extraction"] OR [Full Text: "reference resolution"] AND [Full Text: law] OR [Full Text: legal] OR [Full Text: court] AND [Publication Date: (01/01/1990 TO 05/31/2020)]
IEEE eXplore	(("Full Text & Metadata":"information extraction" AND (law OR legal OR court)) OR ("Full Text & Metadata":"relation extraction" AND (law OR legal OR court)) OR ("Full Text & Metadata":"name entity recognition" AND (law OR legal OR court)) OR ("Full Text & Metadata":" event extraction " AND (law OR legal OR court)) OR ("Full Text & Metadata":" reference resolution" AND (law OR legal OR court)))
Scopus	TITLE-ABS-KEY (("information extraction" OR "relation extraction" OR "named entity recognition" OR "event extraction" OR "reference resolution") AND (law OR legal OR court)) AND PUBYEAR > 2008 AND (LIMIT-TO (LANGUAGE, "English")) AND (LIMIT-TO (SRCTYPE, "p") OR LIMIT-TO (SRCTYPE, "j"))
SpringerLink	'("information extraction" OR "relation extraction" OR "name entity recognition" OR "event extraction" OR "reference resolution") AND (law OR legal OR court)'
ScienceDirect	Find articles with these terms ('information extraction' OR 'relation extraction' OR "named entity recognition" OR "event extraction" OR "reference resolution") AND (law OR legal OR court)Year(s): 1990–2020Article types: review articles, research articles

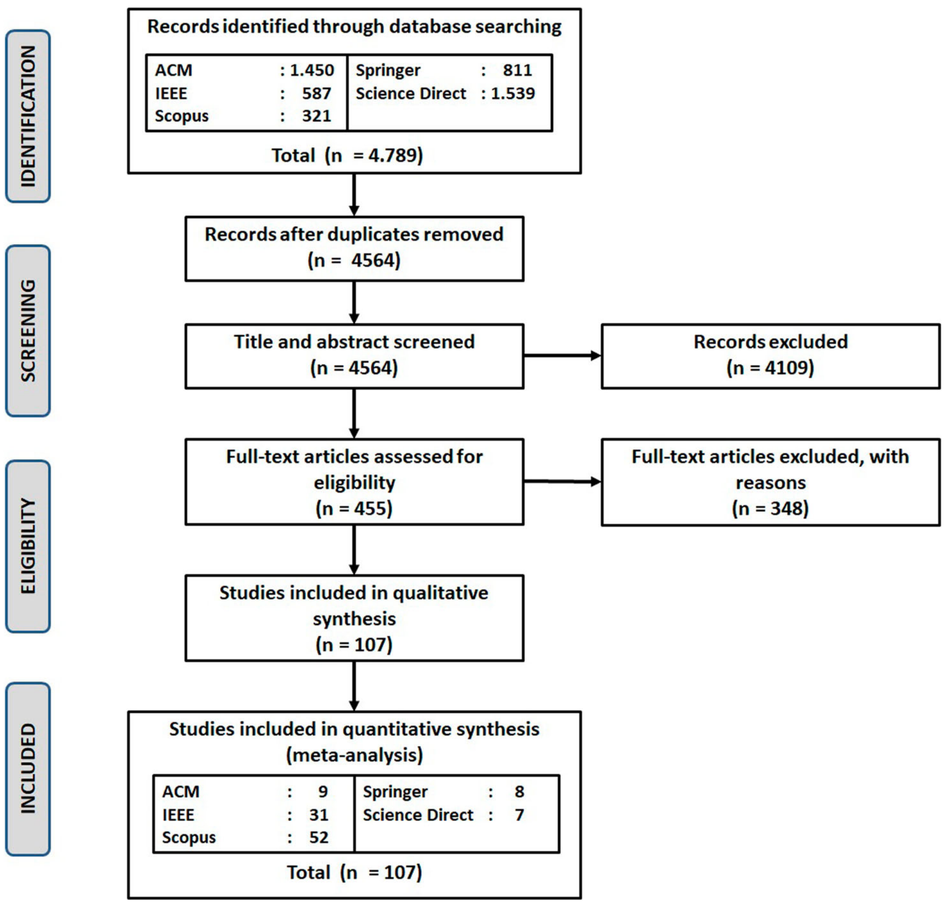

Figure 1. PRISMA flow diagram for the study-selection process.

2.2. Study selection

The study selection process was carried out as shown in Figure 1. During this process, the initial stage encompassed a search performed using keywords in the research database. Then, the search results were identified and filtered using inclusion and exclusion criteria (Table 3) with two filters – that is, title and abstract and full text, respectively. At this stage, 107 studies were obtained for further analysis.

Based on the final list of selected studies, each database offered a percentage of the number of studies selected as shown in Figure 2. Scopus dominated with 48.60%, followed by IEEE with 28. 97% and ACM for 8.41%; studies from SpringerLink and ScienceDirect made up the remaining 7.48% and 6.54%, respectively.

2.3. Data extraction and data synthesis

Data extraction was carried out by completing the data extraction list from each of the selected studies. This data-collection process was designed to answer predetermined research questions. The list of properties used to answer the research questions can be seen in Table 4.

This SR adopted various methodologies – namely, data synthesis related to the research questions – to improve the presentation of legal IE distribution. One of the methods used was the narrative synthesis method, with results visualised in tables and various diagrams.

3. Results

3.1. General overview of the development and relevance of research in legal IE

For the analysis process, this literature study selected 107 studies related to legal IE. The first analysis was carried out by examining the distribution of research on legal IE within the last 30 years. A review of the distribution based on the year of this research can be seen in Figure 3. From Figure 3, it can be seen that the amount of IE legal research, although fluctuating, in general tends to increase. Although in 2007 there was an increase

Table 3. Inclusion and Exclusion Criteria.

Inclusion Criteria	• Research published between January 1990 and May 2020 • Studies discussing the IE process implemented in the legal domain with various legal documents like court decision, patents, etc. • For studies that have dual versions in both a conference proceeding and journal, only the journal version is eligible • For multiple publications of the same study, only the most complete and up-to-date version is eligible
Exclusion Criteria	• Studies that discussed the use of IE but not in the legal domain • Studies related to the legal domain but not focusing on IE • Studies not written in English • Duplicate investigations

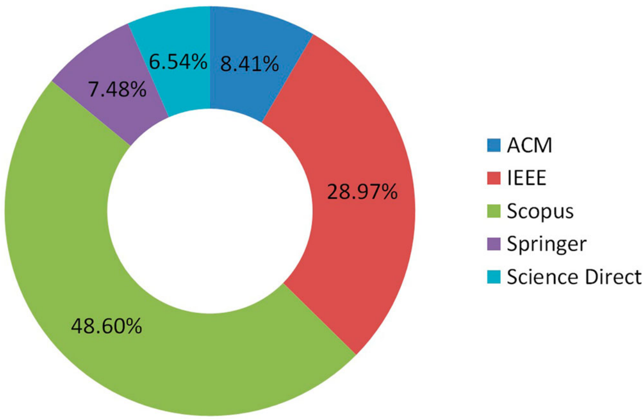

Figure 2. Percentage breakdown of the studies sourced from each database.

in the number of IE legal researches reaching 10 studies and in 2016 there was a decrease to only 2 studies.

The selected studies consisted of a collection of journals and conference proceedings. We found that, if the data from the number of journals and proceedings were to be separated, each of them would still show an increasing trend in the number of papers discussing legal IE from 1997 to 2020, as seen in Figure 4. This condition indicates that research on legal IE is still very relevant today.

To discern the selected journals' reputations, each journal's ranking was assessed on the Scimago Journal Rank (SJR)[6] website. The examination results in the form of a list of the most important journals according to the SJR website and quartile (Q1–Q3) categories of selected legal IE journals are shown in Table 5. The journal *Artificial Intelligence and Law* is the journal that provided the most number of studies to this study. This journal is listed in the best quartile (Q1) for the category of artificial intelligence, with an SJR value of 1.48. This journal explicitly accepts papers that discuss formal or computational models of the legal domain; innovative artificial intelligence systems used in the legal domain; and various interdisciplinary approaches including logic, philosophy, cognitive psychology, linguistics and machine learning.

This SR identified seven journals with the best ranking (Q1 category) based on the SJR among the 16 journals represented in this investigation.

In Table 6, the 16 proceedings included in this study are presented. The International Conference on Artificial Intelligence and Law (ICAIL) was the conference that provided the

Table 4. List of Properties of Data Extraction with the Corresponding Research Question.

Property	Research Questions
Publication and researchers	RQ1
Journal and proceedings	RQ1
Research topics	RQ2
Documents or dataset	RQ3
Legal IE methods	RQ4
Proposed open problem	RQ5

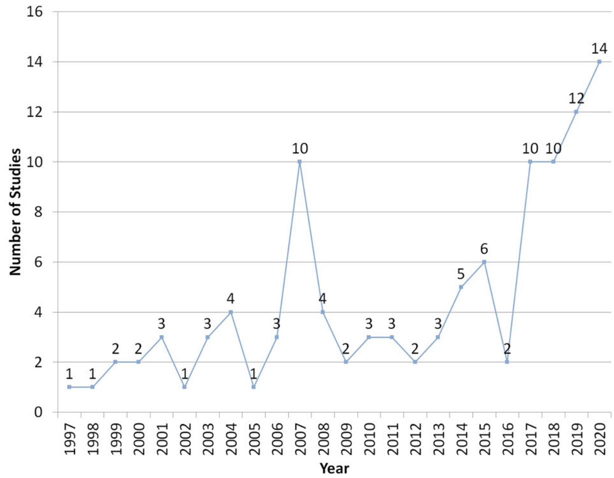

Figure 3. Distribution of selected studies from 1997 to 2020.

greatest number of studies included in this research. ICAIL is a leading international conference that was started in 1987 and which is hosted by the International Association for Artificial Intelligence and Law. It predominantly discusses artificial intelligence research and its application in the legal domain.

The selection of reputable journals and proceedings adhering to the theme of 'legal IE' indicates that this literature review study is appropriate for developing existing 'legal IE' research.

Prominent researchers with an active contribution to legal IE research according to the selected studies can be seen in Figure 5 and include Marie-Francine Moens, Oan Tri Tran, Denis a. de Araujo, Al-Kofahi Khalid, Marijn Schraagen, Caroline Uyttendaele, Alicia Iriberri, Jackson Peter, Ku Chih Hao, and M. Adedjouma. The results of the profile examination of

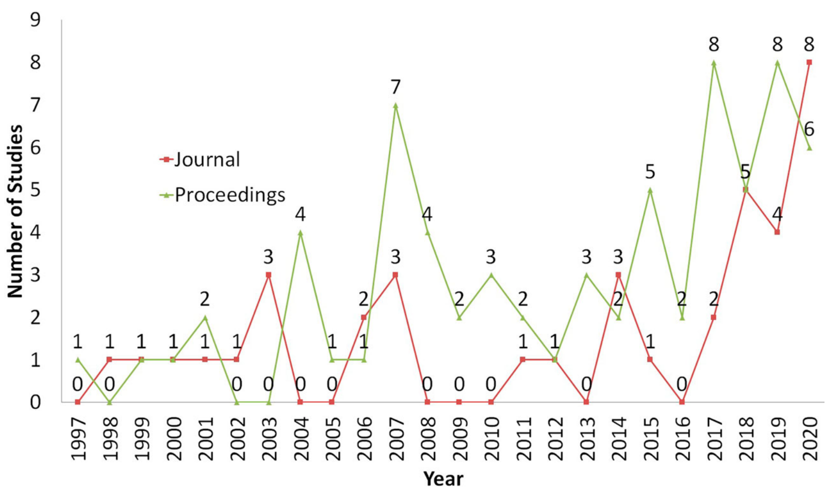

Figure 4. Distribution of journal papers and conference proceedings among the selected studies published from 1990 to 2020.

Table 5. The Essential List of Journals, SJR Values and Quartile Categories.

No	Journal	SJR	Q	Category	No. of Studies
1	*Artificial Intelligence and Law*	1.48	Q1	Artificial intelligence	9
2	*IEEE Transactions on Services Computing*	1.38	Q1	Computer networks and communications	2
3	*Information Processing & Management*	1.19	Q1	Computer science applications	3
4	*Artificial Intelligence*	1.25	Q1	Artificial Intelligent	1
5	*Information Systems Frontiers*	1.01	Q1	Computer networks and Communications	1
6	*IEEE Access*	0.78	Q1	Computer science	1
7	*International Journal of Human-Computer Studies*	0.76	Q1	Engineering	1
8	*Computers & Electrical Engineering*	0.58	Q1	Computer science	1
9	*European Journal of International Law*	0.46	Q1	Law	1
10	*Machine Translation*	0.43	Q1	Linguistics and Language	1
11	*China Communications*	0.49	Q2	Computer networks and communications	1
12	*Information and Communications Technology Law*	0.26	Q2	Law	1
13	*Lingvisticae Investigationes*	0.23	Q2	Linguistics and Language	1
14	*Computational Intelligence*	0.34	Q3	Artificial intelligence	1
15	*Computer Systems Science and Engineering*	0.19	Q3	Computer Science	1
16	*Advances in Intelligent Systems and Computing*	0.18	Q3	Computer science	1

Table 6. The Essential List of Proceedings Found Among the Selected Studies.

No	Proceeding or Conference Title	No. of Studies
1	International Conference on Artificial Intelligence and Law (ICAIL)	11
2	CEUR Workshop Proceedings	6
3	Lecture Notes in Computer Science	3
4	International Conference on Advanced Computer Science and Information Systems (ICACSIS)	2
5	International Conference on Intelligence and Security Informatics	2
6	International Conference on Knowledge Graph (ICKG)	2
7	International Conference on Language Resources and Evaluation, LREC 2004	2
8	International Requirements Engineering Conference (RE)	2
9	Web Information Systems and Applications Conference (WISA)	2
10	International Conference of Computer and Applied Sciences (CAS)	1
11	International Conference of the International Society for Scientometrics and Informetrics	1
12	International Conference on Advanced Computational Intelligence (ICACI)	1
13	International Conference on Advances in ICT for Emerging Regions (ICTer)	1
14	International Conference on Advances in Social Networks Analysis and Mining (ASONAM)	1
15	International Conference on Agents and Artificial Intelligence	1
16	International Conference on Artificial Intelligence Applications and Innovations (IC-AIAI)	1

each of these researchers support that they have published many papers that generally discuss IE including, in particular, that in the legal domain.

Based on the data review and analysis in the General Overview of the Development and Relevance of Research in Legal IE subsection, RQ1 questions regarding the state of the development of research in legal IE can be answered in this subsection. From this description, it is known that the number of papers discussing legal IE for 30 years since 1990 is 107, with a trend that tends to increase every year. From these selected papers, it is known that the journal Artificial Intelligence and Law and the International Conference on Artificial Intelligence and Law (ICAIL) are the journals and conferences that discuss the most about legal IE. Meanwhile, the researcher who has the highest number of IE legal papers is Marie-Francine Moens.

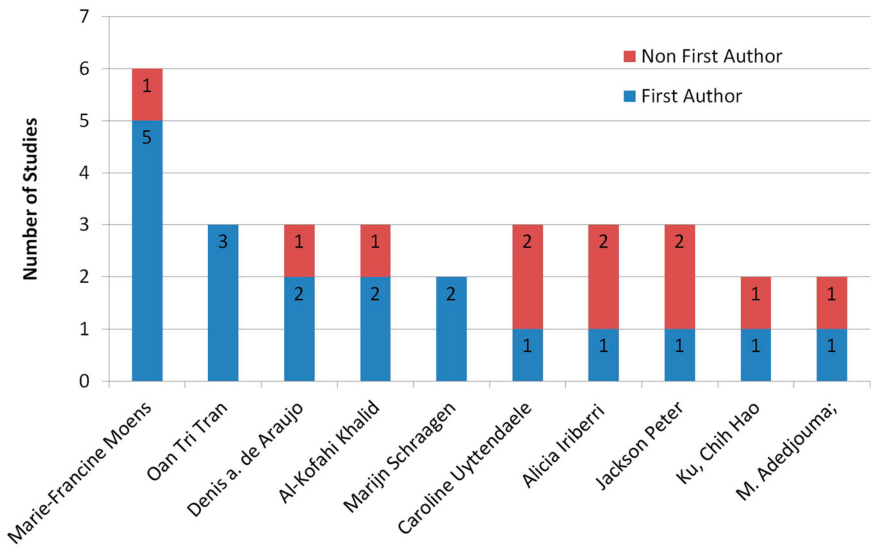

Figure 5. The researchers with the most papers and activity in legal IE research.

3.2. Research topics in legal IE

The purpose of IE is to extract specific pieces of information from the related text, with a set of concepts defined in the extraction scenario (Turmo, Ageno, and Català 2006). IE has several main tasks – that is, identifying entities through a process called named entity recognition (NER), obtaining relationships between entities via relation extraction (RE), finding words referring to the same entity referred to as co-reference resolution (CO), identifying events along with their entities in a process called event extraction (EE) and performing other tasks related to IE such as preprocessing or annotation. This study examines the legal IE task and found several studies that involved NER, CO, RE, EE and other legal IE tasks. NER tasks were the focus of the majority of the research, as can be seen in Figure 6.

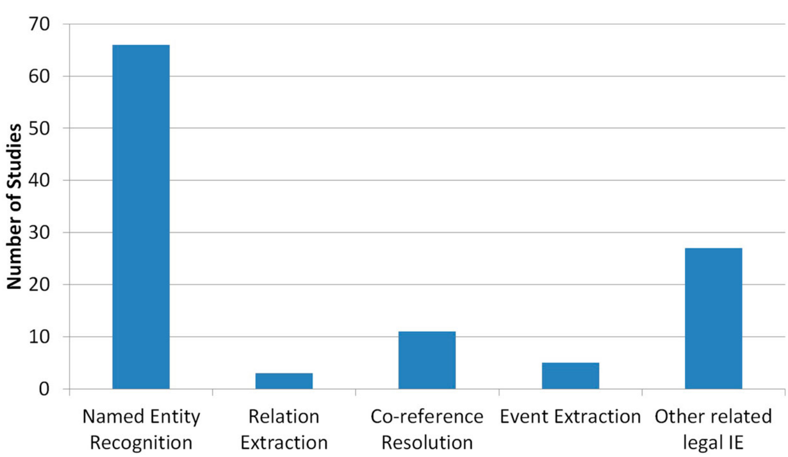

Figure 6. Distribution of research task or topic in legal IE.

The main function of NER is to identify the existence of a predetermined entity, such as organisations, people, place names and others. Additional NER tasks include delineating descriptive information from detected text – for example, extracting the name, sex and other attributes of a person. Studies that discuss NER include papers explaining the NER approach in German language documents from the legal domain. Entities whose extraction totalled 19 semantic classes (e.g. people, judges, lawyers, organisations, regulations, legal norms, etc.) were examined by Leitner, Rehm, and Moreno-Schneider 2019 (Leitner, Rehm, and Moreno-Schneider 2019). Other research performed by Cardellino et al. in 2017 carried out the NER process in contract documents, using and comparing several NER methods (Cardellino et al. 2017). The research of Jasim, Sadiq, and Abdullah 2019 carried out the process of extracting argument entities in the legal domain such as claims, premises and their relationships to provide structured data that could be processed by an argumentation model (Jasim, Sadiq, and Abdullah 2019).

CO requires identification (co-referring) of several entities in common when mentioning the text. Entities can be named, pronounced, nominal or implicit. Several legal IE studies chose CO topics to include research that focuses on reference resolution tasks in the legal domain to obtain references such as that conducted by Thi Tran et al. in 2013 (Tran, Le Nguyen, and Shimazu 2013). Another paper by Thi Tran et al., in 2014 attempted to solve the problem of references to subdocument targets (Tran et al. 2014a). However, research conducted by Van Nam et al. completed the task of detecting references and obtaining references (Huynh et al. 2019).

RE must detect relationships between entities from within the text. Research related to this RE assignment includes that from Schraagen and Bex in 2019, who carried out the RE process of criminal action reports in Dutch (Schraagen and Bex 2019).

EE is tasked with identifying the events of existing entities. Several studies related to the EE task such as: research combining the structured subject event extraction (SEE) method, event extraction (EE) and attribute value pair extraction (AVPE) in the Chinese court decision case conducted by G. Wu et al 2020 (Wu et al. 2020), as well as research another by Schilder F. 2007 regarding Event extraction and temporal reasoning in legal documents (Schilder 2007).

Another related legal IE is another task that is still related to the IE process, especially in the legal domain, such as preprocessing, annotation and IE ontology. Some of the studies that fall into this category include research performed using commercial legal documentation that allows layout rules to be derived and then used to support IE as done by García et al. in 2017 (García-Constantino et al. 2017). Chen et al. in 2015 developed automatic patent annotations using a supervised machine learning algorithm (Chen and Deng 2015). Research that proposed a knowledge extraction approach from legal documents with open information-extraction techniques was carried out without the need for a dataset described in certain documents or specific domains by Kadu in 2020 (Kadu and Zadgaonkar 2020).

Based on the four classic IE tasks – namely, NER, CO, RE and EE – legal IE development is still dominated by NER tasks, with very few CO, EE and RE tasks. Several studies that used more than one IE task included research that combined NER and entity linker tasks, such as that by Cardellino in 2017 (Cardellino et al. 2017). Research that combines NER and summarisation has also been carried out by Kowsrihawat in 2015 (Kowsrihawat and Vateekul 2015). Meanwhile, Cheng performed research in which both NER and CO tasks were carried out (Cheng et al. 2009). Another study that has involved more than one IE task was the 2018 study by Khasianov, who combined NER and RE assignments

(Khasianov et al. 2018). This condition raises the thought that the use of tasks other than NER tasks or the combination of several tasks in legal IE needs further investigation.

The review of data and analysis in the Research Topics in legal IE subsection was used to answer RQ2 regarding topics of research are selected by researchers in legal IE. From the description, the answer is that from several tasks in legal IE, NER is the most frequently discussed topic. In addition to the last five years, there is a trend of research using a combination of several tasks.

3.3. Research datasets in legal IE

In IE research, datasets constitute one crucial factor requiring special attention. This study also conducted a review process of datasets. Based on 107 studies, 53% used court decisions as a dataset, 23% used laws/regulations, 21% used other legal documents and the remaining 4% used open datasets that were already available. The distribution of legal dataset types can be seen in Figure 7.

Further, information about legal dataset types and subtypes and their distribution can be seen in Table 7, which offers information on the lists of legal dataset subtypes that are often used, the number of studies that use them and which research has used these datasets. Besides, there were two open datasets used by the selected study – namely, le NER-Br (Wang et al. 2010) and Multilingual CONLL (Crawley and Wagner 2010).

In IE, language is also a determining factor, especially in the processing of text or documents that generally adhere to the NLP concept. Language differences also affect the complexity, difficulty level and choice of steps and methods in IE processing. In this study, among the 107 studies selected, there were 18 languages identified. If a language is associated with the legal dataset type, we obtained a matrix as seen in Table 8. In Table 8, it is known that the languages of the dataset outside of English that are popular are Chinese, German and Japanese. The variation in treatment associated with language differences is one of the potential directions for further development of legal IE in the future.

Among the included studies, the size of datasets used was very diverse, ranging from the smallest being 10 document records to the highest being 101,562 document records. A depiction was created using a range to facilitate analysis, as shown in Figure 8.

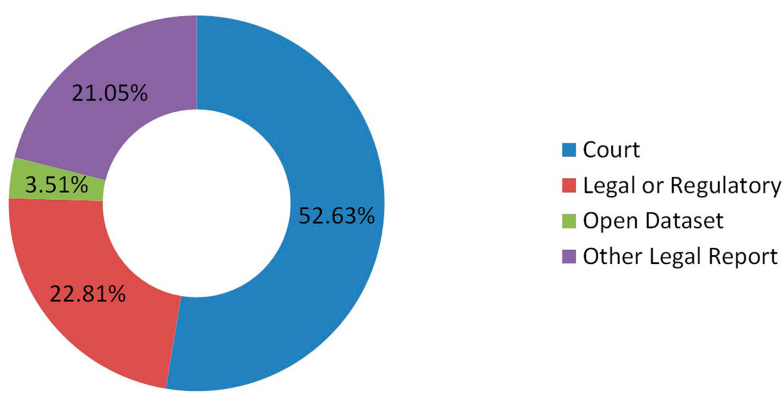

Figure 7. Distribution of legal dataset types.

Table 7. List of Legal Dataset Types and Subtypes with Corresponding References.

No	Legal Dataset Type and Subtype	No. of Studies	Studies
1	**Court**		
	a. Human rights	2	(Cardellino et al. 2017; Medvedeva, Vols, and Wieling 2020)
	b. Commercial	1	(García-Constantino et al. 2017)
	c. Cassation	1	(Jasim, Sadiq, and Abdullah 2019)]
	d. Arbitration	1	(Khasianov et al. 2018)
	e. Appellate/appeal	2	(Fernandes et al. 2020 Sep, Boniol et al. 2020)
	f. Crime/criminal	6	(Solihin and Budi 2019; Iftikhar, Jaffry, and Malik 2019; Ratnayaka et al. 2019; Jin et al. 2020; Mezghanni and Gargouri 2016; Li et al. 2020)
	g. Garnishment	1	(Bordino et al. 2020 Mar 17)
	h. General/unspecific	24	(Leitner, Rehm, and Moreno-Schneider 2019; Kadu and Zadgaonkar 2020; Kowsrihawat and Vateekul 2015; Cheng et al. 2009; Chen et al. 2020; De Araujo et al. 2013; Gaur 2011; Zhuang et al. 2017; Niu and Zeng 2018; Ji et al. 2020a; Delgado 2015; Sharafat, Nasar, and Jaffry 2019; Thomas and Sangeetha 2019; Xi and Zhenxing 2018; Lyte and Branting 2019; Andrei, Sandro, and Victo 2017; Alschner and Charlotin 2018) (Moens 2007; Al-Kofahi, Grom, and Jackson 1999; Uyttendaele, Moens, and Dumortier 1998; de Almeida et al. 2020; Wu et al. 2020; Filtz et al. 2020; Gupta et al. 2018)
2	**Legal or Regulatory**		
	a. Pension	3	(Tran, Le Nguyen, and Shimazu 2013; Tran et al. 2014a; Huynh et al. 2019)
	b. Financial/tax	2	(Plachouras and Leidner 2015; Adedjouma, Sabetzadeh, and Briand 2014)
	c. General/unspecific	11	(Hartadi and Budi 2018; Bach et al. 2019; Janpitak, Sathitwiriyawong, and Pipatthanaudomdee 2019; Kulkarni, Patil, and Shridharan 2017; Navas-Loro 2018; Koboyatshwene, Lefoane, and Narasimhan 2017; Amato et al. 2011; Gianfelice et al. 2013; Boella et al. 2012; Bolioli, Mercatali, and Romano 2004; Lee 2007)
3	Other Legal Report		
	a. Contracts	7	(Chalkidis, Androutsopoulos, and Michos 2017; Annervaz, George, and Sengupta 2016; Nayak and Pasumarthi 2019; Gao, Singh, and Mehra 2012; Glaser, Waltl, and Matthes 2018; Castellanos and Dayal 2004; Gao and Singh 2014)
	b. Notary acts	1	(Buey et al. 2016)
	c. Paten/Intellectual Property Law	4	(Chen and Deng 2015; Loza Mencía 2009; Feng, Chen, and Peng 2011; Fang, Zhang, and Xiao 2007)
	d. Letter of credit	1	(Roychoudhury, Kulkarni, and Bellarykar 2015)
	e. User-submitted crime reports	5	(Schraagen and Bex 2019; Schraagen, Brinkhuis, and Bex 2017; Ku, Iriberri, and Leroy 2008; Iriberri and Leroy 2007; Chih, Iriberri, and Leroy 2008)

An analysis of the dataset's use shows that court decisions dominate with 53%, regulations and laws compose 23%, other legal reports compose 21% and the remainder (open datasets) composes 4%. The small size of open datasets in the legal field is due to the huge effort and cost of collecting and processing legal data. To date, the language used in datasets has remained predominantly English, while Chinese, German and Japanese are the dominant legal data languages after English, respectively. The development of additional datasets using various languages is a challenge in itself for the development of legal IE.

Preparing the dataset should not be separated from the annotation process. From the selected studies, it can be seen that most research adopted a manual annotation process, while only a few have tried using a semi-automatic annotation process (García-Constantino et al. 2017; Bach et al. 2019). Several studies also employed the help of tools in carrying out the annotation process. The list of studies that use annotation tools can be seen in Table 9. The selection of annotation schemes focuses more on the ease of use and the schema's compatibility with the IE method. From the SR, it was found that the most

Table 8. Language of Dataset and Type of Legal Dataset.

		Type of Legal Dataset					
No	Dataset Language	Court	Legal/ Regulatory	Open Dataset	Others Legal Report	Total	Studies
1	English	14	7	2	5	**28**	(Cardellino et al. 2017; García-Constantino et al. 2017; Kadu and Zadgaonkar 2020; Cheng et al. 2009; Wang et al. 2010; Crawley and Wagner 2010; Medvedeva, Vols, and Wieling 2020; Iftikhar, Jaffry, and Malik 2019; Ratnayaka et al. 2019; Bordino et al. 2020 Mar 17; Gaur 2011; Delgado 2015; Sharafat, Nasar, and Jaffry 2019; Thomas and Sangeetha 2019 Nov 25; Lyte and Branting 2019; Andrei, Sandro, and Victo 2017; Alschner and Charlotin 2018; Plachouras and Leidner 2015; Janpitak, Sathitwiriyawong, and Pipatthanaudomdee 2019; Kulkarni, Patil, and Shridharan 2017; Koboyatshwene, Lefoane, and Narasimhan 2017; Amato et al. 2011; Boella et al. 2012; Chalkidis, Androutsopoulos, and Michos 2017; Annervaz, George, and Sengupta 2016; Nayak and Pasumarthi 2019; Gao, Singh, and Mehra 2012; Roychoudhury, Kulkarni, and Bellarykar 2015; Ku, Iriberri, and Leroy 2008; Condon and Miller 2006; Castellanos and Dayal 2004; Li et al. 2007; Iriberri and Leroy 2007; Costa and Neves 2000; Jackson et al. 2003; Moens, Uyttendaele, and Dumortier 1999; Moens 2007; Al-Kofahi et al. 2001; Reiss et al. 2008; Al-Kofahi, Grom, and Jackson 1999; Moens et al. 2007; Chih, Iriberri, and Leroy 2008; McCarty 2007; Bolioli, Mercatali, and Romano 2004; Mazur and Dale 2007; Brüninghaus and Ashley 2001; Holowczak and Adam 1997; Taylor 2004; Moens, Uyttendaele, and Dumortier 2000; Hansen and Selsøe Sørensen 2002; Navarro, Baeza-Yates, and Arcoverde Azevedo 2003; Fang, Zhang, and Xiao 2007; Lee 2007; Martínez, De La Fuente, and Derniame 2003; Ben HacheyClaire 2006; Uyttendaele, Moens, and Dumortier 1998; Marie-Francine 2001; Martínez-González, de la Fuente, and Vicente 2005; Farah and Rousselot 2007; Piskorski et al. 2010; Adedjouma, Sabetzadeh, and Briand 2014; Sannier et al. 2017; de Almeida et al. 2020; Gao and Singh 2014; Wu et al. 2020; Ji et al. 2020a; Finkel and Manning 2008; Schilder 2007; Filtz et al. 2020; Valvoda and Ray 2018; Gupta et al. 2018; Nikolaos et al. 2010)
2	Chinese	7			2	**9**	(Chen and Deng 2015; Jin et al. 2020; Chen et al. 2020; Zhuang et al. 2017; Niu and Zeng 2018; Ji et al. 2020b; Xi and Zhenxing 2018; Feng, Chen, and Peng 2011; Li et al. 2020)

(*Continued*)

Table 8. Continued.

No	Dataset Language	Type of Legal Dataset: Court	Legal/ Regulatory	Open Dataset	Others Legal Report	Total	Studies
3	German	1		1	1	**3**	(Crawley and Wagner 2010; Gaur 2011; Glaser, Waltl, and Matthes 2018)
4	Japanese		3			**4**	(Tran, Le Nguyen, and Shimazu 2013; Tran et al. 2014a; Huynh et al. 2019; Tran et al. 2014b)
5	Dutch					**2**	(Schraagen and Bex 2019; Schraagen, Brinkhuis, and Bex 2017)
6	Indonesian	1	1			**2**	(Hartadi and Budi 2018; Solihin and Budi 2019)
8	Portuguese	2				**2**	(Fernandes et al. 2020 Sep, De Araujo et al. 2013)
9	Spanish		1		1	**2**	(Navas-Loro 2018; Buey et al. 2016)
10	French	1			1	**2**	(Boniol et al. 2020. , Loza Mencía 2009)
11	Arabic	2				**2**	(Jasim, Sadiq, and Abdullah 2019; Mezghanni and Gargouri 2016)
13	Italian		1			**2**	(Gianfelice et al. 2013; Bartolini et al. 2004)
15	Russian	1				**1**	(Khasianov et al. 2018)
16	Thai	1				**1**	(Kowsrihawat and Vateekul 2015)
17	Vietnamese		1			**1**	(Bach et al. 2019)
18	Slovene		1			**1**	(Erjavec and Sárossy 2006)
	Total	**28**	**14**	**3**	**12**		

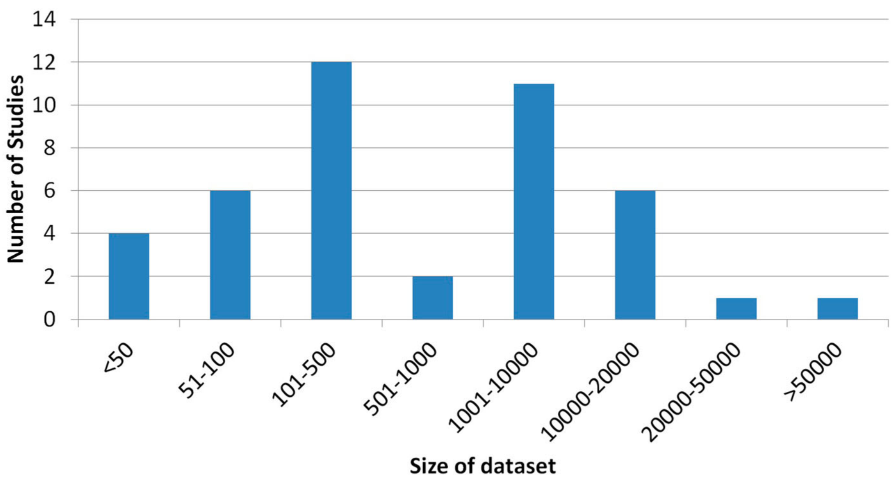

Figure 8. Distribution of the size of record datasets in legal IE.

widely used annotation schemes were XML for rule-based method and IOB for learning-based method.

Feature processing functions to provide various information indicating special conditions and linguistic characteristics are extracted from a set of words to be encoded. Feature processing is one of the essential tasks in IE as it defines the set of features required to predict the targeted entities as accurately as possible.

From the list of the numbers of studies that used features in the IE process in Table 10, it can be seen that the three main features most often used in legal IE research from the

Table 9. List of Studies Employing Tools for the Annotation Process.

No	Annotation Tools	No. of Studies	Studies
1	GATE[a] & JAPE[b]	6	(García-Constantino et al. 2017; Gaur 2011; Janpitak, Sathitwiriyawong, and Pipatthanaudomdee 2019; Ku, Iriberri, and Leroy 2008; Iriberri and Leroy 2007; Adedjouma, Sabetzadeh, and Briand 2014)
2	WEKA[c]	2	(Bordino et al. 2020 Mar 17; Koboyatshwene, Lefoane, and Narasimhan 2017)
3	POWLA[d]	2	(De Araujo et al. 2013; Andrei, Sandro, and Victo 2017)
4	MITRE Annotation Tool (MAT)[e]	1	(Lyte and Branting 2019)

[a]http://gate.ac.uk
[b]https://gate.ac.uk/sale/tao/splitch8.html
[c]http://www.cs.waikato.ac.nz/ml/weka
[d]http://purl.org/powla
[e]https://sourceforge.net/projects/mat-annotation/

selected studies are word shape patterns, part of speech tagging and char n-gram. Meanwhile, the feature groups whose usage is still limited are context, orthographic, lexicon and other customs. Besides, there are still some features that have not been used in the selected studies at all, including suffixes in the morphological group, conjunction in the context group and trigger names in the lexicons group, respectively.

The answer to RQ3 of this SR regarding the most frequently used type of legal datasets for legal IE can be obtained from various descriptions in the subsection of Research Datasets in Legal IE. From the description, it can be seen that court decisions are the most frequently used dataset type. Meanwhile, other important attributes in the dataset include: the most frequently used language is English. The most frequently used annotation schemes are XML for rule-based methods and IOB for learning-based methods. GATE and JAPE are the most frequently used annotation tools. The most used feature in the dataset is morphological. And the most frequently used database size range is between 100 to 500 datasets.

Table 10. List the number of studies that used features in the IE process.

No	Features	No. of Studies	Total
1	**Linguistic**		21
	a. Normalisation	4	
	b. POS	12	
	c. Chunking	4	
	d. Dependency	1	
2	**Orthographic**		4
	a. Capitalisation	2	
	b. Counting	1	
	c. Symbols	1	
3	**Morphological**		23
	a. Suffix and prefix	0	
	b. Char n-grams	10	
	c. Word shape pattern	13	
4	**Context**		2
	a. Windows	2	
	b. Conjunctions	0	
5	**Lexicons**		4
	a. Target names	4	
	b. Trigger names	0	
6	**Other**		6
	a. Customs	6	

3.4. Research methods and improvements in legal IE

For the last ten years, there have been 18 methods used to carry out legal IE processing. The list of methods used in the legal processing of IE is shown in Figure 9. It can be noted that the rule-based is the number one method most widely used, then conditional random fields are in the second position.

From the list of methods most frequently used in legal IE, we took the four most commonly used methods and analysed them based on their distribution across the years. The distribution of the most frequently used methods and by year can be seen in Figure 10. From Figure 10, it can be discerned that rule-based methods have been used almost every year since 2009, followed by SVM while CRF have only been used since 2014 and are exhibiting a growth trend. Meanwhile, BiLSTM have only appeared in use since 2019. This condition is still in line with the fact presented by Chiticariu in 2013, who concluded that commercial IE systems are dominated by the use of rule-based methods as compared with other methods. Besides, academic researchers tend to follow developments in learning methods such as machine learning and deep learning (Chiticariu and Reiss 2013).

The form of research contribution from each study varies. Besides proposing methods and their use and evaluation, the study also often creates and develops a tool or system. For example, the Contract Miner created by Gao et al. in 2012 is a tool developed using a simple but effective unsupervised IE approach to obtain information from contract documents (Gao, Singh, and Mehra 2012). Meanwhile, Khasianov et al. in 2018 conducted research to create an intelligent system named Robot Lawyer to provide information about the legal process required by lawyers and people carrying out legal processes (Khasianov et al. 2018). A complete list of tools developed as a result of the research included in the primary study can be seen in Table 11.

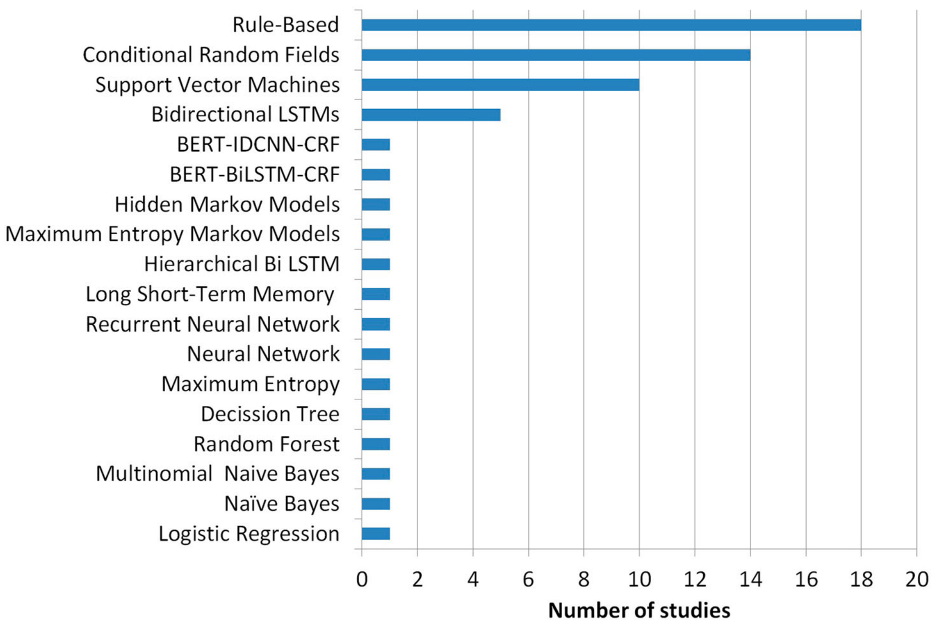

Figure 9. Distribution of research method in legal IE.

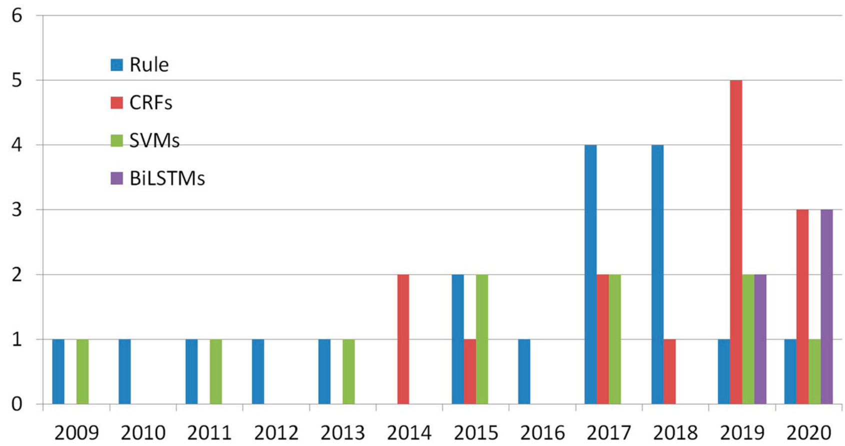

Figure 10. Distribution of the four most commonly research methods used in legal IE.

The review of data and analysis in the subsection on "Research Methods and Improvements in Legal IE" was used to answer RQ4 regarding the kinds of methods are used for legal IE. From this description, it is found that based on the selected paper, rule based is the most often used method for legal IE research. Its development the use of the rule base method in the last five years has tended to decrease and was replaced by the use of learning base method which tends to increase.

4. Discussion

Based on the results of the analysis of the selected research and based on the data, analysis and answers of RQ1 indicate that the number of studies specialising in the legal IE sector is increasing. This finding is in line with the expanding use of information technology, which is generally progressing in many fields, with the legal sector being no exception. Research from McKinsey & Co. provides evidence that 69% of the work in paralegal jobs and 23% of the work in attorney jobs can be automated using artificial intelligence technology in the United States (McKinsey 2014). A graph of the results of McKinsey & Co research can be seen in Figure 11. Research from Levy and Remus also reached a similar conclusion, suggesting that 13% of lawyer work can be automated (Remus and Levy 2016). It can be noted that area of legal IE research is still very relevant to be selected by researcher.

The increase in the use of the latest technology to support or automate various jobs related to legal matters can be demonstrated by the existence of 12 studies that emphasise the development of systems or applications. These studies include research aimed at creating an intelligent system (Lawyer Robot) with the primary function of providing information about the legal process required by lawyers and people who are carrying out legal processes (Khasianov et al. 2018). JudgeDoll is an application used to extract core information automatically and which provides a legal summary of the entire decision to assist judges in summarising legal points quickly, correctly and reliably (Kowsrihawat and Vateekul 2015). The Smart Legal System is a system that extracts entities automatically from the verdict of the Pakistani court system (Sharafat, Nasar, and Jaffry 2019). Finally, research us underway to create a smart system that processes legal texts to

Table 11. List of Tools Proposed by the Selected Studies.

No	Tool Name	Description	Studies
1	Commercial Law Information Extraction based on Layout environment (CLIEL)	An application that serves to complete the documentation for commercial law using the concept of the derivation of layout rules to support the IE process	(García-Constantino et al. 2017)
2	Legal NE Recognizer, Classifier and Linker	This tool is used to recognise a relevant fragment of the text and relate them to a structured knowledge representation that uses the LKIF ontology	(Cardellino et al. 2017)
3	JudgeDoll	An application that is used to extract core information automatically and provide a legal summary of the complete decision to assist judges in summarising legal points quickly, correctly and reliably	(Kowsrihawat and Vateekul 2015)
4	Legal Information Retrieval and Focused Semantic Search (LIRFSS)	This application was used for the semantic search process in Supreme Court decisions, especially in cases of murder, by applying the concepts of IE and Information Retrieval (IR).	(Gaur 2011)
5	Turning Unstructured Texts to Helpful Structure (TRUTHS)	A system that extracts relevant information particularly from the verdicts of criminal cases of the Supreme Court of the Philippines	(Cheng et al. 2009)
6	Legal Mining System (PULMS)	A smart system that processes legal texts to obtain beneficial and prominent information to aid the work of judges and lawyers in the assessment and preparation of cases and to assist people with legal problems	(Iftikhar, Jaffry, and Malik 2019)
7	ANNIC	A GATE plug-in tool to help annotate, visualise and check features with the ability to perform the extraction process in more detail than as seen with ANNIE; this plug-in converts unstructured text into structured data such as tables	(Janpitak, Sathitwiriyawong, and Pipatthanaudomdee 2019)
8	Contract Miner	A simple but effective unsupervised IE approach for discovering service exceptions at the phrase level from a large contract repository	(Gao, Singh, and Mehra 2012)
9	Lawyer Robot	An intelligent system with the primary function of providing information about the legal process required by lawyers and people who are carrying out legal processes	(Khasianov et al. 2018)
10	Joint Method for Litigants Extraction (JMLE)	A method used to make full use of the information from the text itself	(Niu and Zeng 2018)
11	Smart Legal System	A system used to extract entities automatically from the verdicts of the Pakistani court system	(Sharafat, Nasar, and Jaffry 2019)
12	LawORDate	A web service system used to detect legal references with temporal information in Spanish text	(Navas-Loro 2018)
13	History Assistant	A system that uses the concept of information extraction and retrieval to extract decisions from court opinions and retrieve relevant previous cases	(Jackson et al. 2003)
14	Legal Cross Reference Analyzer (LeCA)	a system built to automate the creation of text structure markup, detect & resolution of cross references and analyse & visualise cross references	(Adedjouma, Sabetzadeh, and Briand 2014)
15	FACTS	a system that automatically identifies information from old contracts using	(Castellanos and Dayal 2004)

(*Continued*)

Table 11. Continued.

No	Tool Name	Description	Studies
		information extraction techniques to be used in the management of the contract life cycle effectively and efficiently	

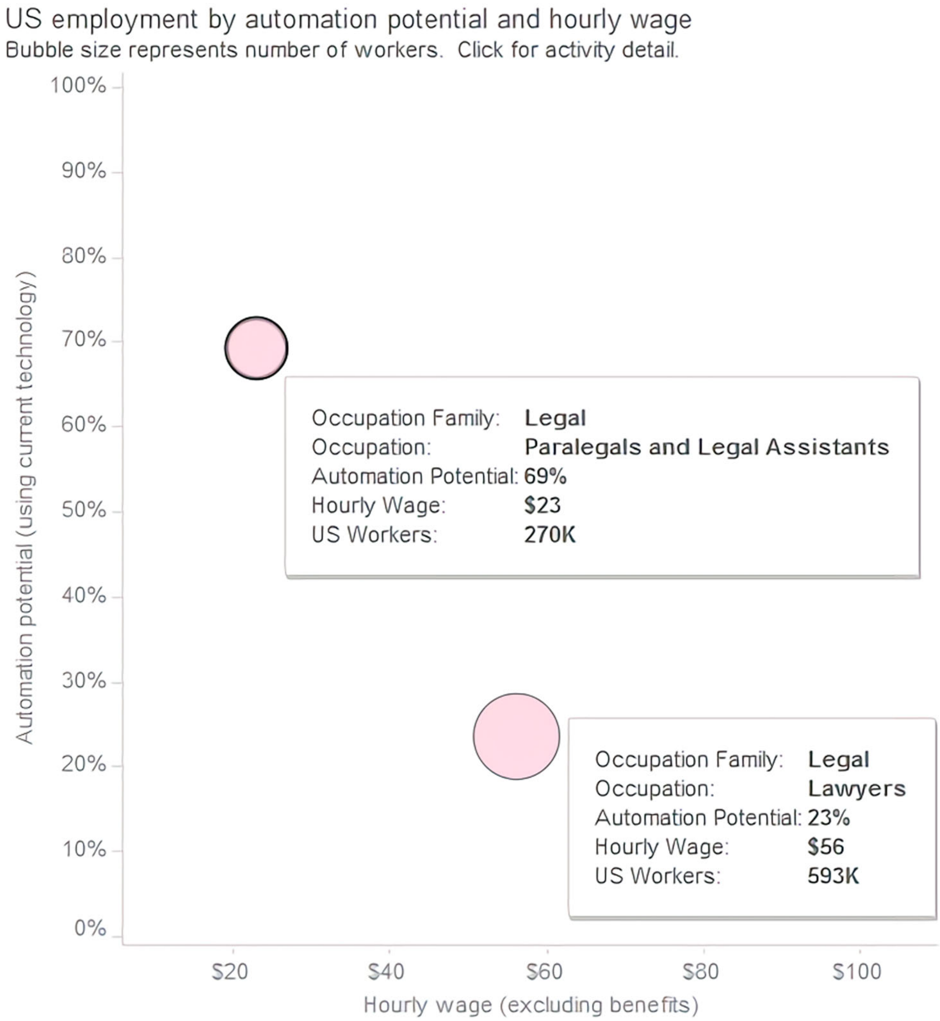

Figure 11. The potential for the automation of the work of lawyers and paralegals in the United States.

Note: https://public.tableau.com/profile/mckinsey.analytics#!/vizhome/AutomationandUSjobs/Technicalpotentialforautomation.

obtain beneficial and prominent information to aid the work of judges and lawyers in the assessment and preparation of cases and to assist people with legal problems, which is called the Legal Mining System (Iftikhar, Jaffry, and Malik 2019).

To answer RQ5 and to make it easier to provide an overview of the findings of this SR, a summary of various items from various processes used in research such as documents,

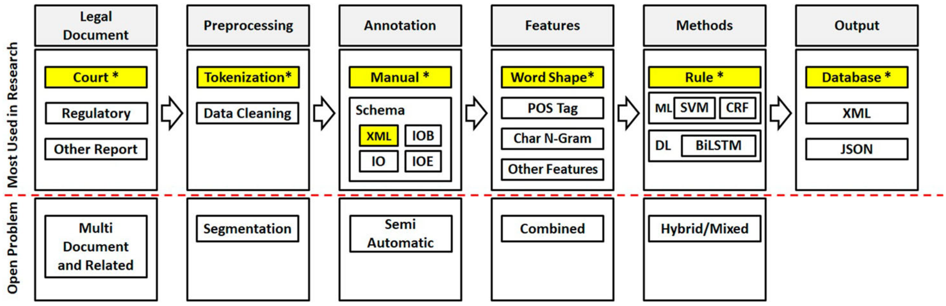

Figure 12. Most common options used in research and the open problem (the yellow box is the item with the highest number of uses).

preprocessing, annotation, features, methods and outputs was made as shown in Figure 12. The items in legal documents, annotations, and features were obtained from the data, analysis, and answers of RQ3. While the The items in preprocessing and method items were obtained from the data, analysis and answers of RQ4 and RQ2. The most widely used item choices in each process are marked with a yellow box to differentiate them from other selected items. Besides, this SR also provides notes on several open problems that can be used as a guideline for developing research in the legal IE sector. In Figure 12, the open problem is placed below the dashed red line. Each open problem will be explained as follows.

There are many types of legal documents and each has a different structure, especially concerning the use of standard language and terms, which can also vary. Each country also has its own standard legal document and uses different languages. This condition makes selecting the optimal legal document type an open problem that can be further worked on. This SR notes that most research focuses predominantly on one particular document, such as criminal court decisions or contracts. Therefore, one of the points of suggestion that can be gleaned is IE for multiple documents that refer to each other and which are closely correlated, such as court decisions, appellate decisions, cassation decisions and judicial review decisions. The example of this proposal poses particular challenges in obtaining entities from each document with a unique structure. Besides, the process of identifying entities and linking each related document or reference is required so that a representation of the complete document linkage for a legal case is formed even though it involves many documents and tiered judicial processes.

Tokenisation is the standard preprocessing approach that is most widely used in the IE process, including in legal IE. This process functions to cut sentences into the smallest meaningful parts. Because legal documents have the characteristics of long and complicated sentences that allow the number of document pages to be large, such as court decision documents, this condition causes the tokenisation process to produce a vast number of tokens. Of course, this condition creates complexity for the IE process, aside from the fact that not all tokens will be used in the IE process. Because it requires a process to divide a document or text functionally and semantically into several segments, some studies refer to this process as segmentation (García-Constantino et al. 2017; Loza Mencía 2009). Which conditions of legal documents are suitable for segmentation and the extent of the effects of segmentation is one of the open problems that can be explored in future research in the legal IE sector, producing improved results and performance.

Preparing annotations is a job that requires a lot of effort and a lot of money. Most of the annotation process is performed manually, with the annotated corpus often divided into intervals of 100 to 500 documents. On the other hand, IE development using training-based methods such as Machine Learning (ML) and Deep Learning (DL) requires a large amount of training data to get better results. This requirement has given rise to various proposals to make the annotation process more comfortable to perform and, at the same time, increase the amount of annotated data. Constantino's 2017 research divides annotated entities into two parts, where one part is done manually and the other part is done automatically using rule-based layout detection (García-Constantino et al. 2017). Another similar effort by Bach in 2019 divided annotation into two parts; first, automatic annotation is performed using regular expression and, second, the annotation is completed manually (Bach et al. 2019). Various trials are required to ascertain the right steps in the semi-automatic annotation process. This condition raises an open problem for legal IE development to help the annotation process run semi-automatically and with better results.

In every IE research, feature selection is always part of the IE process. Selection of the right features can enhance the results. The SR results show that the most frequently used feature is word shape pattern as part of a morphological feature. Studies Jasim, Sadiq, and Abdullah (2019) and Iftikhar, Jaffry, and Malik (2019) compare many features and their combinations. The existence of a large number of features and their combinations is one of the open problems that remains wide open for consideration to improve the results and performance of the IE process.

Hybrid or mixed methods are starting to emerge as the latest method options that can be used for IE processes. IE legal studies (Fernandes et al. 2020; Bach et al. 2019) have already used hybrid methods such as BiLSTM-CRF. The results of these two studies indicate an excellent f1-score in the 95% range. Also, research (Wang et al. 2010) using BiLSTM-CRF and IDCNN-CRF with BERT as pretraining reported an f1-score close to that of the previous study, above 91%. This condition indicates that the hybrid method will likely be more often applied to legal IE research in the future.

Note that some open problems that are answers to the RQ5 of this SR can be considered for future research development processes, either by selecting one open problem or several open problems at once.

5. Conclusions

This SR recognises and analyses developments, topics, datasets and methods in legal IE research. A total of 107 studies published between January 1990 and December 2020 were selected based on predefined inclusion and exclusion criteria. By identifying, assessing and interpreting all the evidence from the research obtained, this literature review intended to answer the research questions that have been determined.

Summary of answers from RQ1 to RQ5 and some observations from this SR concern the advancement of the legal research process in the future, are as follows: first, the trend of the use of the latest technology in general in the legal domain and especially in legal IE is increasing rapidly and has the potential to continue to be developed. Secondly, the exploration of legal IE topics remains dominated by the entity extraction process or NER, so there are still many opportunities to explore various tasks in other IE processes such as EE, CO or preprocessing. Third, there are still many legal dataset types that were not considered in this legal IE research. The use of one language or multiple languages is still limited. The difficulty and the high cost of the legal dataset annotation process are potential and special challenges in conducting a more in-depth research process. Fourth, the choice of methods in legal IE can be broadly grouped as rule-based and learning-based methods (machine learning and deep learning). Each has many variants that can be selected uniquely or can be combined. Besides, it is also possible to create an IE system that can adapt to the rules of various types of legal documents, rules from various countries and different languages. We believe that the analysis of the results in this SR shows that, although the IE research area has been going on for a long time, there is still a lot of work to be completed, especially in legal IE research. The result of this SR is an open problem recommendation that can be used as a guideline in developing legal IE research, as shown in Figure 12.

Abbreviations

The following abbreviations are used in this manuscript:
BiLSTM Bidirectional Long Short Term Memory
CO Co-reference Resolution
CRF Conditional Random Fields
DL Deep Learning
EE Event Extraction
IE Information Extraction
ML Machine Learning
NER Named Entity Recognition
SR Systematic Review
SVM Support Vector Machines
RE Relation Extraction
RQ Research Question

Notes

1. https://dl.acm.org
2. https://ieeexplore.ieee.org
3. https://scopus.com
4. https://springerlink.com
5. https://sciencedirect.com
6. https://www.scimagojr.com

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